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A Combined Incremental Learning Algorithm for Human-Robot Collaboration

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Abstract

Human-Robot Collaboration (HRC) in person-assistance scenarios requires learning natural collaborative behaviors from human demonstrations, rather than repeating fixed routines as in many industrial settings. These behaviors are more complex and non-stationary, needing Incremental and Continual Learning (ICL) methods. In fact, each new user introduces distinct interaction patterns, and the robot's system must continuously interpret multimodal inputs whose cross-modal relationships can drift across users. Existing such learning methods typically rely on either regularization or replay alone, which limits their ability to balance adaptation to new users and retention of prior skills. We propose, in this paper, a hybrid incremental learning algorithm, FLAIR, that combines task-conditioned modulation, task-aware experience replay with dark-knowledge distillation, online Fisher-based importance regularization with retroactive boosting, and multi-head output blending. We train and evaluate FLAIR on a multimodal HRC dataset (joystick commands, robot joint states, and end-effector signals) collected from human operators in shared-autonomy assistance tasks. Results show that FLAIR achieves the best Average Accuracy (ACC) (ACC = 0.690), surpassing existing incremental learning algorithms.

Keywords: human-robot collaboration; incremental learning; multimodal learning; assistive robotics

1. Introduction

HRCs are increasingly deployed in assistive robotics, teleoperation, and rehabilitation, where robots must adapt to diverse users while maintaining reliable behavior over time [1, 2, 3]. Unlike many structured industrial workflows where tasks are well-defined and stationary, person-assistance interactions are natural and user-dependent, with larger variability in intent and motion patterns. In such long-running deployments, new operators are encountered sequentially, and each operator exhibits distinct motor preferences. This requires incremental learning algorithms that must *adapt* to new users without *forgetting* previously learned collaboration behaviors.

This type of learning scenarios involves input multimodality, which in HRC denotes the joint modeling of heterogeneous signals that capture complementary aspects of interaction, such as human intent inputs (e.g., joystick commands), robot proprioceptive state (e.g., joint positions), and task-space motion cues (e.g., end-effector pose/trajectory).

This setting is a clear case of *catastrophic forgetting* [4], where neural networks lose performance on previously learned tasks as training continues on new tasks. In ICL, there are three main types of mitigation strategies: regularization-based approaches, such as Elastic Weight Consolidation (EWC) [5], that penalize updates to parameters important for prior tasks; replay-based approaches, such as Dark Experience Replay++ (DER++) [6], that mix current-task data with buffered past samples; and architecture-based approaches, such as Learning to Prompt (L2P) [7], that allocate task-specific representational capacity to reduce interference.

Each family addresses only a subset of the stability-plasticity trade-off. Regularization-based approaches improve retention but may reduce plasticity for incoming tasks. Replay-based approaches improve memory of prior data but depend strongly on buffer composition and sampling policy, without explicitly constraining parameter drift. Architecture-based approaches reduce interference by task separation, yet their parameter growth can limit scalability as the number of users/tasks increases.

For assistive HRC, this limitation is amplified by multimodal learning demands: the algorithm must preserve cross-modal coordination (human intent signals and robot state dynamics) while adapting to new users. This motivates the need for a stronger incremental learning algorithm tailored to realistic person-assistance settings.

To enable robust adaptation to human intent and gesture variability, we propose FiLM-based Learning with Adaptive Importance and Replay (FLAIR), an integrated hybrid incremental learning algorithm for sequential policy adaptation in HRC. This algorithm is trained on the Human And Robot Multimodal Observations of Natural Interactive Collaboration (HARMONIC) dataset [8], collected from 24 users, imitating assistive eating. The main contributions are:

1. A hybrid algorithm that jointly integrates per-task Feature-wise Linear Modulation (FiLM) conditioning [9], task-aware experience replay with dark knowledge distillation, online Fisher-based importance regularization, retroactive importance boosting (RetroBoost), FiLM warm-start initialization, and forgetting-proportional adaptive replay weighting.
2. Multi-head output blending embedded in the same unified training objective to balance per-task specialization with cross-task knowledge sharing.

2. Related Works

Research in assistive robotics can be grouped into three main areas: human-robot collaboration, shared autonomy, and multimodal learning. Together, these directions show that assistive systems must balance safe, intent-aware interaction with robust fusion of heterogeneous signals, while adapting to diverse users over time.

The authors in [1] introduced hindsight-optimization-based shared autonomy for robotics systems, where autonomous assistance is selected by optimizing expected user outcomes under uncertainty about intent. Their formulation establishes a principled decision-theoretic basis for assistance, but it does not explicitly address long-horizon sequential user adaptation with forgetting constraints.

In [2], the authors presented a deep reinforcement learning formulation of shared autonomy that learns arbitration policies directly from interaction feedback. This line of work improves adaptive control quality, yet it mainly optimizes task performance per training regime rather than preserving previously acquired user-specific behaviors across many users.

The work presented in [3] surveyed autonomy levels in physical human-robot interaction and highlighted design trade-offs between safety, transparency, and adaptation. Importantly, the survey emphasizes that adaptation mechanisms in assistive interaction must remain reliable and interpretable, which becomes harder when the user population grows over time.

In [10], the authors presented a review of shared control in physical human-robot interaction, with a structured analysis of three core components: intent detection, arbitration between human and robot commands, and communication/feedback to the user. Their work clarifies design trade-offs and evaluation criteria for assistive shared autonomy, and provides a conceptual foundation for building reliable human-in-the-loop controllers.

More recently, [11] proposed Human-Robot Gym as a benchmarking environment for reinforcement learning in collaborative settings, enabling standardized comparison across HRC control strategies. Such benchmarks strengthen reproducibility, but they also reveal that most current evaluations still focus on per-task adaptation quality rather than long-term incremental retention.

Collectively, these works show that person-assistance HRC increasingly depends on intent-aware control, multimodal representation learning, and robust adaptation under user variability. However, an important gap remains: existing approaches are rarely designed as integrated incremental-learning systems that jointly optimize adaptation to new users and retention of prior user-specific motor collaboration patterns.

Building on this state of the art, we propose FLAIR, a unified incremental learning algorithm that addresses catastrophic forgetting in sequential multimodal HRC while keeping memory usage practical and performance strong across seen users.

3. Methodology

3.1. Problem Formulation

We consider the task-incremental learning setting. A sequence of N tasks $\{T_1, \ldots, T_N\}$ arrives one at a time, where each task T_i corresponds to a distinct human operator in a natural assistive interaction. For task T_i, the agent observes multimodal inputs (joystick commands, joint positions, and end-effector positions; observation $o \in \mathbb{R}^{d_o}$) and must predict the corresponding joint velocity commands (action $a \in \mathbb{R}^{d_a}$). The goal is to learn a policy π_θ that performs well on *all* tasks after sequentially training on each, while maintaining robust cross-modal representations as user dynamics evolve.

3.2. Proposed Architecture: FLAIR

FLAIR uses a shared Multilayer Perceptron (MLP) backbone with per-task FiLM conditioning layers. The architecture consists of:

$$h^{(l)} = \text{Dropout}\left(\gamma_t^{(l)} \cdot \text{ReLU}\left(W^{(l)} h^{(l-1)} + b^{(l)}\right) + \beta_t^{(l)}\right) \tag{1}$$

where $W^{(l)}, b^{(l)}$ are *shared* backbone weights and $\gamma_t^{(l)}, \beta_t^{(l)}$ are *task-specific* FiLM parameters. The output is produced by blending a shared head with a task-specific head:

$$\hat{a} = (1 - \alpha_h) \cdot f_{\text{shared}}(h^{(L)}) + \alpha_h \cdot f_t(h^{(L)}) \tag{2}$$

where $\alpha_h = 0.3$ is the head blending coefficient.

After training on task t, its FiLM parameters and task head are *frozen*, preventing retroactive modification and ensuring stability.

Fig. 1 illustrates FLAIR's design: data flows through FiLM-conditioned shared layers that are modulated by task-specific parameters (γ_t,β_t), then through two output heads (shared and per-task) that are adaptively blended. Three auxiliary mechanisms support incremental learning: a task-aware replay buffer with adaptive weighting, Fisher-based importance regularization with retroactive boosting, and FiLM warm-start initialization. Frozen components after training are marked with *snowflake* icons.

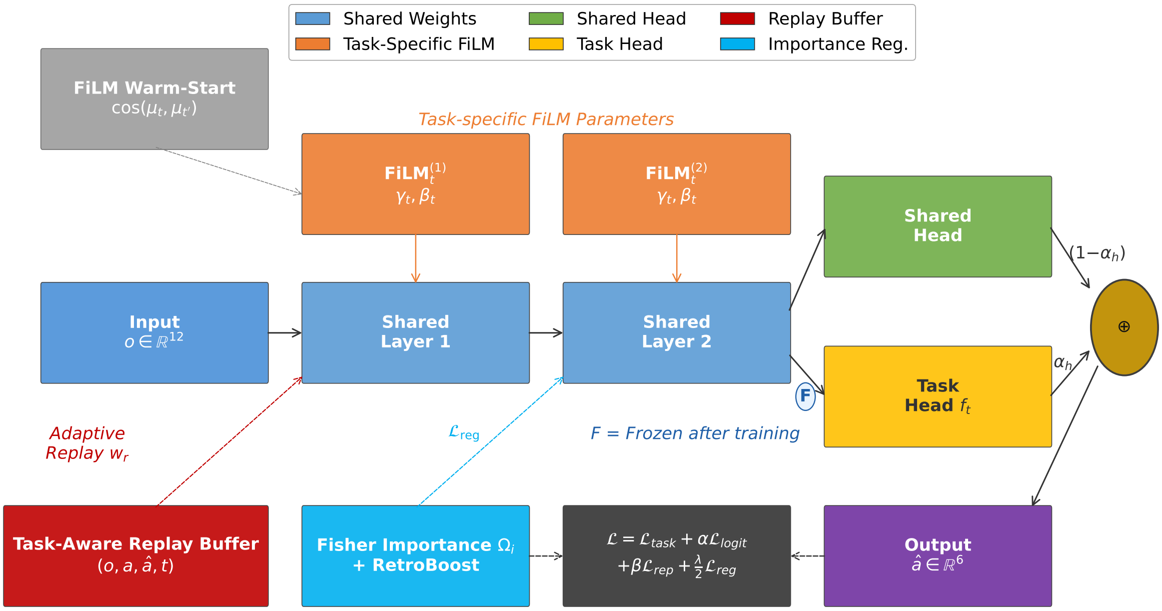

Fig. 1. FLAIR architecture and auxiliary mechanisms

3.2.1. *Training Objective*

The total training loss for task t combines four terms:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha\mathcal{L}_{\text{logit}} + \beta\mathcal{L}_{\text{replay}} + \frac{\lambda}{2}\mathcal{L}_{\text{reg}} \tag{3}$$

where:

Task loss: Standard Mean Squared Error (MSE) on the current task's data:

$$\mathcal{L}_{\text{task}} = \frac{1}{|B_t|}\sum_{(o,a)\in B_t} ||\pi_\theta(o; t) - a||^2 \tag{4}$$

Logit distillation loss: For replayed samples (o_r, t_r) from the buffer, we encourage consistency with stored logits $\hat{a}_r$:

$$\mathcal{L}_{\text{logit}} = \frac{1}{|B_r|}\sum_{(o_r,t_r)\in B_r} w_r||\pi_\theta(o_r; t_r) - \hat{a}_r||^2 \tag{5}$$

where w_r is the adaptive replay weight (Section 3.2.5). **Replay target loss:** Direct MSE on replayed ground-truth actions:

$$\mathcal{L}_{\text{replay}} = \frac{1}{|B_r|}\sum_{(o_r,a_r,t_r)\in B_r} w_r||\pi_\theta(o_r; t_r) - a_r||^2 \tag{6}$$

Importance regularization: Fisher-based penalty on shared parameters:

$$\mathcal{L}_{\text{reg}} = \sum_i \Omega_i(\theta_i - \theta_i^*)^2 \tag{7}$$

where Ω_i is the cumulative importance and θ_i^* is the anchor value after the previous task.

3.2.2. Task-Aware Replay Buffer

Unlike standard replay buffers, FLAIR stores the *task identity* alongside each sample: $(o, a, \hat{a}, t)$. During replay, each sample is routed through its original task's FiLM parameters, ensuring the correct feature modulation is applied. This is critical because shared backbone features are modulated differently for each task; replaying a sample without the correct FiLM routing would produce incorrect gradients.

3.2.3. Retroactive Importance Boosting (RetroBoost)

Standard online EWC accumulates importance as $\Omega^{(t)} = \gamma\Omega^{(t-1)} + F^{(t)}$, where $F^{(t)}$ is the Fisher information for task t. We augment this with a retroactive boost:

$$\Omega^{(t)} \leftarrow (1 + 0.1 \cdot |\text{completed tasks}|) \cdot \Omega^{(t)} \tag{8}$$

The intuition is that as more tasks are learned, the shared backbone becomes more valuable and should be protected more aggressively. This growing multiplier naturally increases regularization strength as the risk of catastrophic forgetting grows.

3.2.4. FiLM Warm-Start

When a new task t arrives, we compute the cosine similarity between its input statistics (mean and standard deviation of observations) and those of all previously learned tasks. The FiLM parameters for t are initialized by cloning those of the most similar task:

$$t^* = \arg\max_{t' \in \text{completed}} \cos\left([\mu_t; \sigma_t], [\mu_{t'}; \sigma_{t'}]\right) \tag{9}$$

$$\gamma_t^{(l)} \leftarrow \gamma_{t^*}^{(l)}, \quad \beta_t^{(l)} \leftarrow \beta_{t^*}^{(l)} \tag{10}$$

This accelerates convergence for similar users and provides a better starting point than the identity transform.

3.2.5. Adaptive Replay Weighting

Standard replay samples uniformly from the buffer. FLAIR tracks each task's peak R^2 and current R^2. For each replayed sample from task t_r, its weight is:

$$w_r = 1 + 5 \cdot \max\left(0, R^2_{\text{peak}}(t_r) - R^2_{\text{current}}(t_r)\right) \tag{11}$$

Tasks experiencing greater forgetting (larger gap between peak and current performance) receive proportionally more replay emphasis.

3.2.6. Algorithmic Visualization

Algorithm 1 summarizes the full FLAIR incremental training pipeline, including FiLM warm-start, task-aware replay, adaptive replay weighting, and Fisher-based regularization with RetroBoost.

3.3. Used dataset

We use the HARMONIC dataset [8], a multimodal human-robot interaction dataset collected from participants controlling a Kinova MICO robotic arm positioned on a table, with a food tray placed in front of the arm (Fig 2). Each participant holds a 3D joystick controller to teleoperate the robot during natural feeding tasks. The shared-autonomy system records data synchronously at the robot's control frequency: joystick inputs from the human operator, proprioceptive joint angles and velocities from the robot, and end-effector pose. This multimodal measurement captures both the human intent (via joystick) and the robot state, reflecting the natural diversity of control strategies and individual behavioral differences across users. Such heterogeneous interaction patterns form the basis for studying incremental learning and user-specific adaptation.

Algorithm 1 FLAIR Incremental Training Procedure

Input: task sequence $\{T_1, \ldots, T_N\}$, shared parameters θ, replay buffer $\mathcal{M}$
Initialize: task statistics store, importance weights $\Omega \leftarrow 0$, anchor parameters $\theta^* \leftarrow \theta$
for $t = 1$ to N **do**
  if $t > 1$ **then**
    Compute task similarity statistics and warm-start (γ_t, β_t) from most similar prior task
  else
    Initialize (γ_t, β_t) to identity
  end if
  for each training mini-batch B_t from task T_t **do**
    Sample replay mini-batch B_r from $\mathcal{M}$ with stored task IDs
    Route each replay sample through its original task FiLM branch
    Compute adaptive replay weights $w_r \leftarrow 1 + 5\max(0, R^2_{\text{peak}} - R^2_{\text{current}})$
    Compute total loss:
      $\mathcal{L} \leftarrow \mathcal{L}_{\text{task}} + \alpha\mathcal{L}_{\text{logit}} + \beta\mathcal{L}_{\text{replay}} + \frac{\lambda}{2}\mathcal{L}_{\text{reg}}$
    Update θ, (γ_t, β_t), and task head parameters via backpropagation
    Store sampled current-task examples and logits $(o, a, \hat{a}, t)$ in $\mathcal{M}$
  end for
  Estimate Fisher information $F^{(t)}$ on task T_t
  Update cumulative importance: $\Omega \leftarrow \gamma\Omega + F^{(t)}$
  Apply RetroBoost: $\Omega \leftarrow (1 + 0.1 \cdot |\text{completed tasks}|)\Omega$
  Set anchor parameters: $\theta^* \leftarrow \theta$
  Freeze task-specific FiLM parameters and task head for task t
end for
Output: trained FLAIR model with shared backbone, frozen task-specific FiLM modules, and replay memory

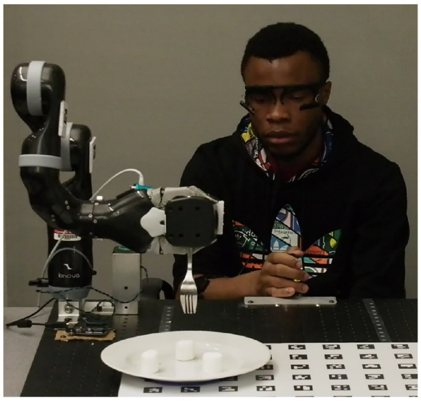

Fig. 2. Experimental setup: shared autonomy assistive eating task [8]

3.3.1. Feature Extraction

From each trial, we extract three input modalities recorded at the robot's control frequency:

Joystick commands (3D): user input axes (x, y, z); **Joint positions** (6D): Kinova MICO 6-DOF joint angles; **End-effector position** (3D): Cartesian coordinates (x, y, z).

These are concatenated into an observation vector $o \in \mathbb{R}^{12}$. The target action $a \in \mathbb{R}^6$ consists of the six joint velocity commands.

Table 1. Per-task dataset statistics (10 participants from HARMONIC).

Task	ID	Trials	Samples	Train	Val	Test
T_0	P100	19	30390	21273	4558	4558
T_1	P101	20	38281	26796	5742	5742
T_2	P102	20	47961	33572	7194	7194
T_3	P103	20	24207	16944	3631	3631
T_4	P104	14	25524	17866	3828	3828
T_5	P105	10	15807	11064	2371	2371
T_6	P106	19	45334	31733	6800	6800
T_7	P107	17	20586	14410	3087	3087
T_8	P108	14	14645	10251	2196	2196
T_9	P109	20	30113	21079	4516	4516
Total		173	292848	204988	41923	41923

3.3.2. Preprocessing

Raw sensor data undergoes three preprocessing steps: (1) *Outlier clipping*: joint position values exceeding |50| rad are replaced by the per-column median of non-outlier values, correcting occasional sensor glitches. (2) *NaN removal*: timesteps containing any NaN value in observations or targets are dropped. (3) *Global normalization*: after loading all participants, observations and actions are independently z-score normalized using their respective global means and standard deviations computed across all training splits.

3.3.3. Task Construction

Each participant constitutes one task in the incremental learning sequence ($N = 10$ tasks). Per-participant data is randomly shuffled (seed = 42) and split 70%/15%/15% for train/validation/test. Table 1 details the per-task statistics.

The model is trained on one participant at a time in order $T_0, \ldots, T_9$ and evaluated on *all* seen participants after training on each task.

3.4. Baselines and Implementation Details

We compare FLAIR against four established methods:

Joint Training (upper bound): retrains from scratch on all accumulated data for each new task, representing the best achievable performance without memory constraints. **Online EWC** [12]: regularization-based method with running Fisher matrix ($\lambda = 10000$, $\gamma = 0.99$). **A-GEM** [13]: memory-based gradient projection method (256 samples per task). **DER++** [6]: dark-knowledge replay with logit distillation ($\alpha = \beta = 0.5$, buffer = 10000).

All methods share the same MLP backbone (2 hidden layers, 256 units each, ReLU activation, dropout = 0.1) and training protocol (Adam optimizer, lr = 10^{-3}, batch size 256, max 100 epochs with patience-10 early stopping). FLAIR-specific hyperparameters are: replay buffer capacity = 10000, replay weights $\alpha = \beta = 0.5$, importance $\lambda = 5000$, Fisher samples = 2000, FiLM learning rate = 5×10^{-3}, head blend $\alpha_h = 0.3$.

4. Results and Discussion

4.1. Comparison

To jointly measure the key requirements of our setting, standard incremental learning metrics will be used: final utility (**ACC**), retention under sequential updates (**F** and **BWT**), transfer to future tasks (**FWT**), and practical deployability in assistive robotics (memory footprint and training time) [14]. In this work, utility is computed with the coefficient of determination R^2, which quantifies the proportion of variance in joint-velocity targets explained by the model predictions.

Table 2 presents the comparison between different methods: FLAIR, DER++, Online EWC, A-GEM, and Joint Training. On the 10-task benchmark, FLAIR achieves the best average accuracy (ACC = 0.690), lowest forgetting (F = 0.017), and strongest backward transfer (BWT = −0.017) among incremental methods, with memory overhead of only 1.3 MB. Remarkably, FLAIR also outperforms Joint Training (the offline upper bound with full data retraining)

Table 2. Incremental Learning Performance (10-task HRC benchmark)

Method	ACC↑	F↓	BWT↑	FWT↑	Mem (MB)	Time (s)
Joint Training	0.644	0.037	−0.004	+0.013	1664.2	7648
Online EWC	0.415	0.211	−0.211	+0.083	0.5	124
A-GEM	0.420	0.376	−0.376	−0.049	0.2	167
DER++	0.612	0.148	−0.148	**+0.172**	0.9	154
FLAIR (ours)	**0.690**	**0.017**	**−0.017**	+0.118	1.3	555

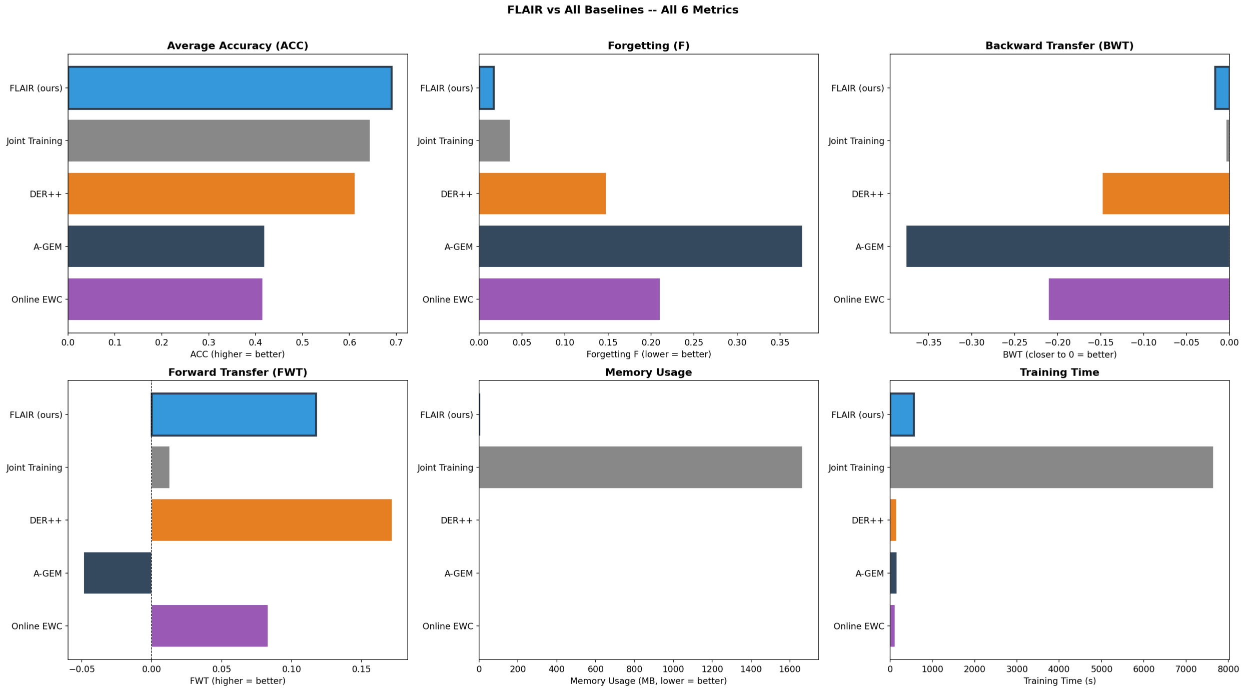

Fig. 3. Metric comparison across all methods

in final accuracy (0.690 vs. 0.644). Higher values are better for ACC, BWT, and FWT; lower values are better for F and memory.

Compared with DER++, FLAIR improves ACC by +12.7% relative and reduces forgetting by 8.8×, with a modest memory increase (+0.37 MB). Compared with Online EWC and A-GEM, gains are larger in both final accuracy and retention stability. Fig. 3 displays all six incremental learning metrics (ACC, F, BWT, FWT, memory, time) across all methods, with FLAIR (blue) ranking first in ACC, F, and BWT while maintaining competitive FWT and practical computational costs.

4.2. Forgetting and Retention Dynamics

Fig. 4 (left) tracks R^2 on Task 0 as subsequent tasks are learned sequentially. FLAIR maintains 94.0% of its initial Task 0 performance after learning 9 additional tasks (R^2: 0.768 → 0.722), compared to DER++ (81.9%), Online EWC (46.0%), and A-GEM (45.8%). Joint Training shows 114.5% retention since it retrains on all data. The right panel shows average R^2 across all seen tasks over time, confirming FLAIR's stable learning trajectory.

4.3. Per-Task Robustness

FLAIR's R^2 coefficients (performance on each task immediately after training) shows consistent quality across users: mean = 0.7047, min = 0.5770 (Task 6), max = 0.8266 (Task 2). This spread indicates heterogeneity in operator

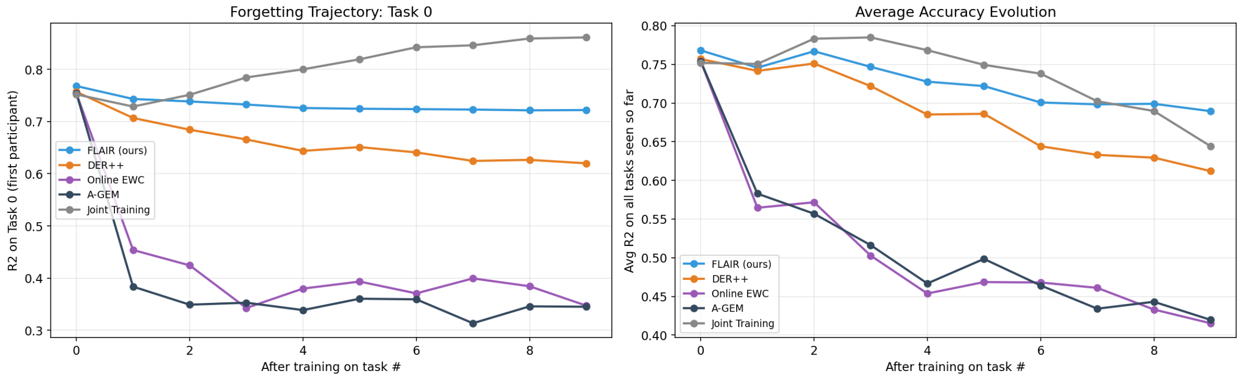

Fig. 4. Retention dynamics across tasks

behavior, while the consistently positive and high per-task scores suggest that the model does not over-specialize to only a subset of users.

4.4. Discussion and Practical Implications

Joint Training is conventionally viewed as an upper bound in incremental learning, yet the observed gains of FLAIR indicate that this assumption can fail when user-specific adaptation is required: FiLM-based task conditioning provides per-user feature specialization that a single shared representation cannot capture, replay and importance regularization jointly stabilize previously learned behaviors and reduce overfitting to recent users, and warm-start initialization transfers useful structure from similar prior users to accelerate adaptation. From a deployment perspective, this combination is attractive for assistive systems because it preserves near-zero forgetting with only 1.3 MB of additional memory, avoiding the impractical cost of storing full historical datasets; although training is slower than lightweight replay-only baselines, the increased computational cost is offset by substantially better reliability across previously seen users. Overall, these results support integrated hybrid incremental designs as a more suitable strategy than single-family methods for long-horizon, multimodal HRC adaptation.

5. Conclusion

Assistive human-robot collaboration requires models that can continuously adapt to new users while preserving previously learned behaviors under multimodal variability and strict memory constraints. In this setting, relying on only one incremental learning strategy is often insufficient to balance plasticity and stability.

To address this challenge, FLAIR is introduced as a unified hybrid incremental learning algorithm built to maintain a stable long-term balance between plasticity and memory retention, so the system can keep learning in realistic assistive HRC without losing earlier collaboration skills.

On a sequential multimodal benchmark, FLAIR achieves better results over other methods, outperforming established ICL baselines and the Joint Training upper bound. These results indicate that combining complementary mechanisms in a single framework is an effective strategy for robust user adaptation in assistive robotics.

Further validation on additional benchmarks would strengthen these findings and help define future research directions.

References

[1] S. Javdani, S. S. Srinivasa, J. A. Bagnell, Shared autonomy via hindsight optimization, in: Robotics: Science and Systems (RSS), 2015, dOI: https://doi.org/10.15607/RSS.2015.XI.032.

[2] S. Reddy, A. D. Dragan, S. Levine, Shared autonomy via deep reinforcement learning, in: Robotics: Science and Systems (RSS), 2018, dOI: https://doi.org/10.15607/RSS.2018.XIV.005.

[3] M. Selvaggio, M. Cognetti, S. Nikolaidis, S. Ivaldi, B. Siciliano, Autonomy in physical human-robot interaction: A brief survey, IEEE Robotics and Automation Letters 6 (4) (2021) 7989–7996, dOI: https://doi.org/10.1109/LRA.2021.3100603.

[4] M. De Lange, R. Aljundi, M. Masana, S. Parisot, X. Jia, A. Leonardis, G. Slabaugh, T. Tuytelaars, A continual learning survey: Defying forgetting in classification tasks, IEEE Transactions on Pattern Analysis and Machine Intelligence 44 (12) (2022) 9771–9788, dOI: https://doi.org/10.1109/TPAMI.2021.3057446.

[5] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, R. Hadsell, Overcoming catastrophic forgetting in neural networks, Proceedings of the National Academy of Sciences 114 (13) (2017) 3521–3526, dOI: https://doi.org/10.1073/pnas.1611835114.

[6] P. Buzzega, M. Boschini, A. Porrello, D. Abati, S. Calderara, Dark experience for general continual learning: A strong, simple baseline, in: Advances in Neural Information Processing Systems (NeurIPS), 2020, dOI: https://doi.org/10.48550/arXiv.2004.07211.

[7] Z. Wang, Z. Zhang, C.-Y. Lee, H. Zhang, R. Sun, X. Ren, G. Su, V. Perot, J. Dy, T. Pfister, Learning to prompt for continual learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, dOI: https://doi.org/10.1109/CVPR52688.2022.00024.

[8] B. A. Newman, R. M. Aronson, S. S. Srinivasa, K. Kitani, H. Admoni, HARMONIC: A multimodal dataset of assistive human-robot collaboration, arXiv preprint arXiv:1807.11154DOI: https://doi.org/10.48550/arXiv.1807.11154 (2022).

[9] E. Perez, F. Strub, H. de Vries, V. Dumoulin, A. Courville, FiLM: Visual reasoning with a general conditioning layer, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2018, dOI: https://doi.org/10.1609/aaai.v32i1.11671.

[10] D. P. Losey, C. G. McDonald, E. Battaglia, M. K. O'Malley, A review of intent detection, arbitration, and communication aspects of shared control for physical human-robot interaction, in: Applied Mechanics Reviews, 2018, dOI: https://doi.org/10.1115/1.4039145.

[11] J. Thumm, F. Trost, M. Althoff, Human-robot gym: Benchmarking reinforcement learning in human-robot collaboration, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, dOI: https://doi.org/10.1109/ICRA57147.2024.10610705.

[12] J. Schwarz, W. Czarnecki, J. Luketina, A. Grabska-Barwinska, Y. W. Teh, R. Pascanu, R. Hadsell, Progress & compress: A scalable framework for continual learning, in: Proceedings of the International Conference on Machine Learning (ICML), 2018, dOI: https://doi.org/10.48550/arXiv.1805.06370.

[13] A. Chaudhry, M. Ranzato, M. Rohrbach, M. Elhoseiny, Efficient lifelong learning with A-GEM, in: Proceedings of the International Conference on Learning Representations (ICLR), 2019, dOI: https://doi.org/10.48550/arXiv.1812.00420.

[14] D. Lopez-Paz, M. Ranzato, Gradient episodic memory for continual learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2017, dOI: https://doi.org/10.48550/arXiv.1706.08840.